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Roll No

EC-4004-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Communication Systems

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume any missing data.

1. a) Find the Fourier transform of a rectangular pulse 2 seconds long with a magnitude of 10 volts.
b) State and prove linearity and frequency convolution property of Fourier transform.
2. a) Determine the signal energy and signal power for
 - i) $f(t) = e^{-3|t|}$ and
 - ii) $f(t) = e^{-3t}$
b) With the help of neat sketches, explain the generation and detection of SSB-SC signal.
3. a) An amplitude modulated signal is represented by
$$\phi_{AM}(t) = 10 \cos(2\pi \cdot 10^6 t) + 5 \cos(2\pi \cdot 10^6 t) \cos(2\pi \cdot 10^3 t) + 2 \cos(2\pi \cdot 10^6 t) \cos(4\pi \cdot 10^3 t)$$
 volts.
Find the various frequency components present, corresponding modulation indices and the bandwidth.
b) Explain the necessity of modulation.

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4.
 - a) Draw and explain Super heterodyne Receiver.
 - b) Calculate the figure of merit in DSB-SC system.
5.
 - a) With the help of block diagram, describe the functioning of a broadcast FM receiver clearly bringing out the functional aspects of each one of the building blocks.
 - b) What is an Image frequency? How is image frequency rejection achieved?
6.
 - a) Draw the block diagram of producing FM from PM. Also explain and write its advantages.
 - b) Compare direct and indirect method of generation.
7.
 - a) Write down the classification of noise.
 - b) Explain the effect of noise on AM and FM receivers.
8. Write short note on any two:
 - a) Comparison between AGC and AFC
 - b) Discriminator method of FM detection
 - c) Synchronous generation and detection

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